

Sound

Every sound starts with a vibration, like the quivering of a guitar string. The vibration squeezes and stretches the air between the vibrating object and your ear. This is a sound wave.



Silent space

Sound can travel through solids, liquids, and gases, but it can't travel where there is no matter. There is no sound in space because there is no air.



Sound waves travel through air like a wave along a coiled spring.

How hearing works

When a sound reaches your ears, it makes your eardrums vibrate. The vibrations are passed to your inner ear through tiny bones. From here, nerves send messages to your brain that allow you to recognize the sound.

Measuring sound

Loudness is measured in decibels.



Rustling leaves make a sound of only 10 decibels.



Whispering measures about 20 decibels.



City traffic reaches approximately 85 decibels.



Drums make a sound of around 105 decibels.



Road-drills measure about 110 decibels.



A lion's roar has been recorded at 114 decibels.



Fireworks can measure 120 decibels or more.



Jet engines sometimes hit 140 decibels.